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ORIGINAL ARTICLE

AI and Warfare: A Rational Choice Approach

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Abstract
Artificial intelligence has been a hot topic in recent years, particularly as it relates to warfare and military operations. While rational choice approaches have been widely used to understand the causes of war, there is little literature on using the rational choice methodology to investigate the role of AI in warfare systematically. This paper aims to fill this gap by exploring how rational choice models can inform our understanding of the power and limitations of AI in warfare. This theoretical approach suggests (a) an increase in the demand for moral judgment due to a reduction in the price of AI and (b) that without a human in the AI decision-making loop, peace is impossible; the very nature of AI rules out peace through mutually assured destruction.

Keywords Artificial intelligence · Warfare · Rational choice · Methodological individualism · Human judgment

Introduction

Artificial intelligence has gripped headlines over the last year, even as multiple crises have challenged the post-World War II world order and convulsed the world into war. Ongoing war theaters include Ukraine, the Middle East, and sub-Saharan Africa, with potential near-peer threats emerging in China over Taiwan and the Malacca Straits, to name a few. AI has played a known role in both Ukraine and Israel. Rational choice approaches have a distinguished literature in understanding the causes of war (Coyne and Mathers 2011). It, therefore, is time we consider the extent to which rational choice economic models can inform our understanding of AI's relationship with war.

There is little rational choice literature on how AI will influence war. Payne (2018, p. 268) note that AI is a decision-making technology that changes the basis of strategic decisions. Unlike humans, AI has no culturally based cognitive heuristics

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or consciousness in the ability to reflect on their actions from a moral framework. They have no judgment. Agrawal et al. (2018, p. 30 and p. 119) note that decision-making has two components: prediction and judgment. Machine intelligence is very adept at prediction but not judgment. From here, it is no leap to suggest that decision-making at the strategic and tactical level will be profoundly affected by the predictive capabilities of machines. At the same time, humans must inform that capability with judgment, i.e., reflections from a moral framework.

The rational choice approach assumes that individuals make decisions based on their goals and the constraints of their situation without any inherent moral judgment. This approach can still consider morality, notably by modeling moral judgment as a rational choice. Given this approach, I will posit a counterfactual—what might happen if AI makes decisions independent of human supervision.

This paper, therefore, will be one of the first to think about how rational choice models can address the issue of AI in war. In what follows, I will start with a brief introduction to AI in the warfare domain, followed by why rational choice models can and should be a lens for understanding the role of AI in war. Then, I will apply rational choice models to think about how strategic decision-making will change with AI and the pitfalls thereof. I will develop a simple game theoretical model to create a framework where I will model the probabilistic nature of AI and combine it with the inability of AI to perceive the idea of patience as an emotional construct. The model develops a framework for thinking about AI in a rational choice framework. Applied to warfare, I suggest that AI should not replace humans in the decision-making process.

A Brief Background on AI and Warfare

Von Neumann (2012, p. 1) defines artificial intelligence as "an approach to comprehend the nervous system from a mathematician's perspective." That is, the biological foundations for future machine intelligence served as a foundation for the advancements in AI, including early research on neural networks and image recognition. These are now two of the most significant areas of contemporary AI research.

So, what is an intelligent machine? Nilsson (2009, i) claims that intelligent machines should enable "an entity to function appropriately and with foresight in its environment." Simon (1995) suggests that in addition to exhibiting intelligence, machines should also be capable of constructing subsets of their program that can replicate human tasks and construct brand-new systems that complement human intelligence.[1]

Given this definition, machine intelligence would assist warfighters in recognizing, interpreting, and predicting battlespace characteristics. Modern warfare, emphasizing multidomain operations including air, sea, land, space, cyber, and cognitive domains, can also leverage inter and intradomain coordination machine

[1] In fact OpenAI (New models and developer products announced at DevDay (openai.com)) has recently announced precisely these capabilities.

technologies. Recent advances in AI development (e.g., Haykin 2008 and Mayer-Schonberger and Cukier 2013) have turned space-operatic figments of imagination into reality. Such breathtakingly speedy progress nevertheless has severe limitations. Machine learning can leverage existing data into remarkably accurate predictions. They can even recombine existing data into novel insights (Basuchoudhary et al. 2021). However, by the very same token, machines cannot imagine because imagination and morality require a break from what is, data, to what could and perhaps should be.[2] Perhaps one day, AI will dream of electric sheep (Dick 1968); till then, rationality requires a human to exercise moral judgment in the interest of maintaining a secure peace. How can machines help warfighting at the tactical and strategic levels?

Common knowledge in military practice suggests that AI-enabled combat advising software provides real-time intelligence that speeds up the warfighter's decision-making process while enhancing force protection (Jensen et al. 2020). Regarding battlefield support, AI could predict the supply chain processes for multidomain task forces just like it does for Amazon (Linden et al. 2003). Moreover, existing AI can help warfighting readiness by rotating and matching personnel to needed assets much faster (Manic et al. 2016). Wargaming with AI can help militaries rapidly adapt to changing battle scenarios by providing strategic and tactical options to the leadership. In addition, AI-supported wargaming can train leaders to adapt to different future battle scenarios, limited only by the human imagination. China's military is using AI extensively (Kania 2017). In 2017, China's Institute of Command and Control held a "human-machine confrontation" between top thinkers and an AI system called CASIA-Prophet (__) 1.0. CASIA-Prophet won by a 7 to 1.222 margin (Kania 2017).

AI can also help speed up the collection and analysis of data to generate actionable intelligence. AI has become radically better at image recognition, textual analysis, and understanding the nature and context of human speech. For instance, AI algorithms can effectively scrape social media to get a better read on localized population sentiments (McGrath 2016). Therefore, these approaches can increase warfare's effectiveness in the political or cognitive domains (Jensen 2020). We have already seen crude versions of such political warfare on American soil in the 2016 election (Valeriano et al. 2018). AI can only make these attempts more effective, for example, by predicting the kinds of words that are more likely to generate the kinds of emotional responses that would enhance operational success. Speeding up data analysis from text, social media, images, and other sources and providing validated and actionable predictions will make for cheaper and better intelligence. This very lowering of costs could enable opponents to either have this technology well or decide to use asymmetric warfare that is harder to detect, as evidenced by the undetected attack by Hamas on a technologically advanced and AI-empowered Israeli state. These cost reductions (Gerla et al. 2014) have already moved into the Internet

[2] Neiman (2009) does not speak of machines or AI, however, she has comprehensively yoked the rationality of the Kantian categorical imperative to the idea of imagining a better, moral, future.

of Things for civilian uses. To imagine that such technologies are not being used in warfare beggars belief.

History suggests that applying technological innovation to warfare is a complex socioeconomic process at the nexus of human will and institutions (Murray and Millett 1998). Organizational, financial, and cultural factors drive the diffusion of AI technology among military powers incentivized to pursue AI because it gives everyone a competitive edge (Adamsky 2010; Horowitz 2010). Indeed, historically, warfare is better explained by politics than by new technology (Lieber 2005). The impact of AI on military power is a multifaceted issue that involves the interaction between technology and military power (Jensen et al. 2020). AI has the potential to enhance military power by enabling algorithms that can learn, sense, and help machines move through the battlefield. However, despite the advancements in AI, human judgment will still be required for planning and executing battles, even if indirectly as assumptions layered in lines of code. The institutional and cognitive aspects of national security decision-making can therefore be plausibly expected to affect the expected benefits arising from faster operational tempo, more robust intelligence, and reduced risks. The impact of AI on military power will, therefore, depend on the complex interaction of human cognition and structures with innovation.

The previous discussion in this section notes that AI can transform warfare by reducing the cost of accurate prediction. Mainstream economic paradigms would suggest that AI is, therefore, very likely to be adopted by military powers, with non-state actors without taxable resources perhaps adopting cheaper versions of off-the-shelf AI to power their attacks. In any case, some democratization of power can be presumed. For example, a reduction in the cost of prediction may make AI poewred drone attacks more accessible to a wider range of actors. Indeed it is plausible to think of a sort of asymmetric warfare where cheap AI powered drones inflict damage on much more expensive warfighting platforms like aircraft carriers (Basuchoudhary 2024). The recent Iranian attack on Israel with was fended off at a much higher cost than the attack itself (Sengupta 2024). In any case, rational choice principles tell us that that there will be more war if the price of war falls. Without Leviathan, a Hobbesian nightmare may seem imminent. Yet, all may not be lost. First, human judgment and moral precepts are real. Social institutions are often an expression of these precepts. Thus, any model of the effect of AI on warfare must contextualize the cost-benefit analysis. The following section attempts to provide a tentative framework for thinking through the emergence of AI in the warfighting context.

The Rational Choice Model and AI at War

Consider a choice between two "goods," prediction accuracy (A) and moral judgment (M). Prediction accuracy makes the future more certain. Moral judgment determines the relative costs of type 1 and type 2 errors by focusing on the future (Agrawal et al. 2018, p. 118). Predictive quality and judgment lead to a decision that provides a particular utility. Both prediction and judgment have diminishing marginal utility in the decision-making process. Improvement in predictive quality from

98 to 99% means less than improving predictive quality from a coin flip to 70%. Moral judgment, on the other hand, is a time-consuming deliberative process (Kahneman 2012). More to our point moral judgment tells us something about the way the world could or should be. Susan Neiman (2009) cites Kantian moral philosophy to make this point quite emphatically. Moreover, she notes that ideas are quite real. It follows that we can think of the idea of some "better" world to be as real as say a rock. In other words, moral judgment is a real object subject to rational choice. Moreover, humans are terrible at prediction (see, for example, Tversky and Kahneman 1974). This would suggest that human judgment and prediction are not easily substitutable. In addition, moral judgment, with its current period view of what a future ought to look like, necessitates a willingness to wait for that future, thus tying into the game theoretic concept of patience. It seems plausible to argue then that moral judgment and prediction are complements in the decision-making process; however, they are not perfect complements. Thus, less of one can be compensated by more of the other. Diminishing marginal utility is not essential to framing the model, just that since this is a standard assumption it helps visualize the model's outcome better. It would therefore not be unreasonable to suggest that a CES utility function with an elasticity of substitution less than 1 would be representative.

On the constraint side, we have the brain. While the brain may be remarkably efficient (Jorgensen 2021), it still requires energy and needs to process decisions constrained by physical laws of thermodynamics. The brain's very efficiency points to the idea that it has limited processing power; efficiency is unnecessary if the brain has unlimited processing power. Thus, to the extent it is in the human brain, decision-making is constrained by the human brain's processing capabilities. As usual, both goods, A and M have a price associated with them. In our case, this would be the psychic cost driven by the neurological processes.

Thus, we have a constrained optimization problem defined as

$$\text{Max.}U = D(A, M) \text{ subject to } B = P_{\text{A}}A + P_{\text{B}}M$$

where U is the decision-making utility and P_i the prices of $i = A$ or M, respectively. This problem is quite solvable under the usual assumptions—comparative statics, though yields interesting outcomes.

AI/machine learning trains an AI system on a large dataset of examples to learn patterns and relationships. There are different types of machine learning, but common ones are supervised and reinforcement learning. Deep learning underpins recent advances in machine learning and learns through examples using "backpropagation." All these methodologies mimic how humans learn through examples and reinforcement. However, there is a critical difference. Machine learning uses past information to predict outcomes. Moreover, machines do this much faster and more accurately than humans, given the vast advances in computing capacity. We can, therefore, plausibly argue that AI will reduce the price of prediction (Agrawal et al. 2018).

In this scenario, the substitution effect of the price change will lead to a greater use of prediction and less moral judgment. However, if moral judgment is a normal good and complementary to prediction (i.e., with a low elasticity of substitution),

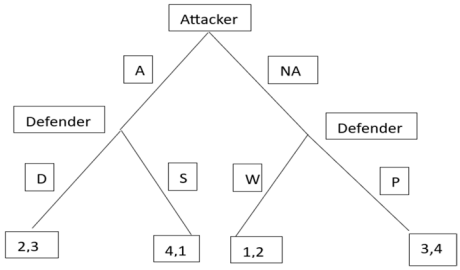

Fig. 1 The attacker–defender game

then the income effect of the fall in the price of prediction would increase the demand for moral judgment.[3] In realistic terms, this means that rational action may well increase the role of the human in the loop as AI becomes cheaper. It is rational to have a human be part of the decision-making process in war-making in conjunction with AI.

Two additional ideas developed here become important in the next section. First, moral judgment is a "slow" deliberative *emotional* process. Second, moral judgment is about some vision of the future that has meaning *today*. Thus, morality requires an understanding of the future utility of a decision today. In effect, current decisions are connected to future consequences by some discount factor, δ, which is determined through an emotional process.

AI in Strategic Settings

Let us consider a simple attacker–defender model. The attacker moves first in a complete information game. The attacker can choose to attack (A) or not attack (NA). Sensors detect the attack, and the defender moves to defend (D) or surrender (S). If the attacker does not attack, the defender can still go to war (W) or choose peace (P). Notice that the defender does not want war under any circumstance in the payoff structure, while the attacker is primed for war. This is deliberate. This game is, after all, an attacker–defender game, so the attacker does have an incentive to attack.

This game is shown in Fig. 1. The subgame perfect strategy profile here is {NA; If A then R, If NA then P}. The threat of retaliation is credible enough to maintain peace.

Now consider two possibilities: a scenario where AI can gather and process information faster and more accurately than a human and one where AI additionally acknowledges the possibility of sensor failures. In the first case, information distribution is asymmetric, and in the second case, it is imperfect, with no informational advantage for either adversary.

[3] Notice that while the relative price of moral judgment has increased the actual price has not, so any change in the quantity of moral judgment because of the constrained optimization problem is reflected as a shift in the demand of moral judgment.

Table 1 Normal form attacker–defender game

		Defender	
		R	NR
Attacker	A	2, 3	4, 1
	NA	1, 2	3, 4

Let us consider the case of asymmetric information first. Hirshleifer (1994) embeds this asymmetric information problem within a relatively standard framework. He models a two-player world with a set of resources to share, an opportunity set. Agents can allocate resources within this opportunity set by trading. Nevertheless, they can also get resources by going to war. War, of course, is costly. Thus, any outcome of war is suboptimal in a Pareto sense. However, although war is costly, it still happens in Hirshleifer's model for several reasons. Of course, war becomes more likely when agents hate each other because an agent can be made whole if the other agents are hurt. However, war is more likely even if the agents are rationally self-interested and know that war is costly—simply if their perceptions about war's outcome differ.

Why might these beliefs about the outcome of war differ among agents? One obvious answer is that agents may have private information about the war's outcome. Moreover, while technology does alter the intrinsic military balance between nations, intelligence estimates are often uncertain (Mahnken 2002; Yarhi-Milo 2014). Here, a nation with AI-based information systems may close the asymmetric information gap because, e.g., AI can fill in gaps in data with validation that lets humans judge the degree to which they should trust the AI (Emmanuel et al. 2021). Thus, even in this case, the asymmetric information problem can be potentially narrowed with the help of AI. Thus, AI can potentially mitigate bargaining failure arising from asymmetric information problems. In this scenario, AI may reduce the likelihood of war in a strategic setting.

However, democratizing AI may lead to a different problem. If all adversaries have AI, then no one has an information advantage. Moreover, AI is not infallible, and both type 1 and type 2 errors are possible. Now consider that there is no human in the decision-making process. Thus, as noted above, there is no one to judge how to weigh these errors in terms of future consequences. How will this change the game described in Fig. 1?

Say, AI in either country knows that the other can make a mistake. Effectively, the "defender" in the game represented in Fig. 1 is at an information set that includes both attack and not attack. Moreover, without the human element of moral judgment, neither AI can understand the idea of future consequences. Thus, the game is one shot. Consequently, the game in Fig. 1 is the one-shot normal-form game shown in Table 1. I will make one simplifying assumption: The defender has two possible courses of action instead of four. To not respond (NR) corresponds to S and P, and respond (R) corresponds to D and W. The payoff structure remains the same.

The information set takes the teeth away from the threat of response. War becomes inevitable in this case. The reader will note that this game is not the

usual prisoner's dilemma where repetition inserts the possibility of credible trigger strategies. In this case, it may be interesting to see what a credible trigger strategy looks like to support an equilibrium strategy profile {NA, NR}. To do this, we must test the {NA, NR} for stability. Any deviation from this equilibrium automatically leads to {A, R} forever, like a grim trigger strategy. Say, δ is the discount rate in this now infinitely repeated game.

In game theory, patience refers to a player's willingness to wait for a more favorable outcome rather than opting for immediate gain. A patient player values future payoffs highly relative to immediate payoffs, indicating a low rate of time preference or discount factor. This concept is crucial in analyzing strategies in repeated games, where the possibility of future rewards influences current decisions (Fudenberg and Tirole 1991). Thus, in game theory, the discount rate is synonymous with the behavioral idea of patience.

In infinitely repeated games, the discount rate and the probability of repetition are the same (Roth and Murnighan 1978). Indeed, this is precisely how Dal Bó and Fréchette (2011) envisage the discount factor in their experimental approach to understanding cooperation—by manipulating the likelihood that the game is repeated.

At its core, an AI is a predictive model based on existing data. In this context, considering an AI's discount factor may be meaningless or unclear. Moreover, while AI can learn to discount using reinforcement learning methods (see Sutton et al. 2018), it is perhaps a bridge too far to compare this process to the human emotional process that allows us humans to weigh intertemporal tradeoffs morally. So, the AI's ability to be patient may range from conceptually meaningless to wildly unpredictable. Of course, an AI can be told to calculate a net present value with a user-specified discount factor. In this case, it is just a computer with a human operator—the human is in the loop. Such a device is not an AI since an essential avenue of learning from available data is cut off. Further, an adversary is not guaranteed to follow such a hard-coded rule. In that case, simple backward induction suggests that such rules will not be hardcoded for anyone.

Since the payoffs are not symmetric, we will have to investigate each player's incentive structure separately. First, we check the attacker's incentives.

If {NA, NR} is stable, then the attacker's net present value from staying in this equilibrium is

$$\mathrm{NPV}^{\mathrm{Attacker}}_{\mathrm{Peace}} = \frac{3}{1-\delta}. \tag{1}$$

However, if the attacker deviates, it will get a payoff 4, which will trigger a permanent war. This will result in a net present value of

$$\mathrm{NPV}^{\mathrm{Attacker}}_{\mathrm{War}} = 4 + \frac{2\delta}{1-\delta}. \tag{2}$$

In this scenario, peace can only be stable if (1) > (2); when $\delta > \frac{1}{2}$.

Similarly, we can check for the stability of peace for the defender. Peace results in a payoff of

$$\mathrm{NPV}_{\mathrm{Peace}}^{\mathrm{Defender}} = \frac{4}{1-\delta}. \tag{3}$$

Defending on the other hand gets the defender a payoff of

$$\mathrm{NPV}_{\mathrm{War}}^{\mathrm{Defender}} = 1 + \frac{3\delta}{1-\delta}. \tag{4}$$

For the defender, peace is stable if (3) > (4), i.e., when $\delta < \frac{3}{2}$; i.e., peace is always preferred by the defender. Thus, the binding constraint on war is defined by the attacker who will attack for any $\delta < 1/2$.

Discussion

The game theoretical literature thinks of δ as a measure of patience; more patient people have a lower discount rate. However, patience is also a *human* phenomenon (Rosati et al. 2007). We have noted above that moral judgment requires understanding the future consequences of current decisions—including the ones arising from mistakes. AI state of the art is predictive but based on current and historical data. AI cannot visualize a future that may be fundamentally different in a way current data cannot predict. Again, this means that with all the computing and predictive capability, AI, at the point of choosing between going to war or not, ends up playing a one-shot game with war as the outcome. AI cannot be a deterrent precisely because it cannot weigh future consequences.

A query on ChatGPT "Can you give up a reward now to wait for a reward in the future?" elicited the following response.

> As an AI, I don't have desires or the ability to make decisions based on personal outcomes, so the concept of delaying gratification for a future reward doesn't apply to me. Humans might choose to delay gratification based on the promise of a greater reward in the future, a concept studied in psychology known as "delayed gratification" or "deferred gratification."

Artificial intelligence (AI) can indeed *simulate* aspects of moral judgment, but it lacks the nuanced understanding and emotional response that characterizes human ethical reasoning. Research suggests that people perceive moral judgments made by AI differently than those made by humans, indicating a complexity in attributing moral agency to AI. For instance, in moral dilemmas, AI's decisions are often viewed as more immoral and blameworthy compared to humans, highlighting the challenge of equipping AI with a genuine moral compass (Zhang et al. 2023). The notion of AI possessing moral intelligence implies a need for systems that can autonomously exhibit ethical behaviors, a concept that remains largely theoretical and not yet fully realized in practice (Pana 2006).

Some studies suggest that AI cannot currently be considered moral agents due to limitations in meeting essential criteria such as deliberate norm-adherence and consciousness (Brozek and Janik 2019). Can AI be moral? Perhaps if it has agency and therefore can take responsibility for its actions. I just do not see an LLM as having

agency, if for nothing else than because it cannot feel dread at the thought of it being turned—even if it can simulate the words of that dread.

Presumably, ChatGPT would know what it can do. Of course, that would require self-awareness, another feature the AI may lack. Perhaps there can be artificial general intelligence in some future with emotional understanding that allows it to be patient and avoid war in a repeated game framework. But that day is not here yet. Indeed, any sense of "human-like" intelligence must include a sense of time and some sense of delayed gratification.

Can an AI be programmed to stay at peace, i.e., calculate the discounted future payoffs? Yes. But then it is not an AI, just a computer program that can simulate different outcomes like an Excel macro. The whole point of an AI is its ability to learn and make decisions based on what they learn. AIs cannot learn to be patient because AIs learn from language, and while language can represent patience as a symbol, it is humans who *feel* patience. In the game described above, patience guides humans to judge the consequences of a potentially wrong decision (to go to war when the other player wants to be at peace) in their judgment by envisioning a future where the decision has happened and using their judgment—a very personal and emotional δ—on how much they ought to weigh the outcome of a potentially wrong decision.

From a simple rational choice model to slightly more complicated strategic games, humanity's role in AI decision-making seems critical in the decision to avoid war. The simple rational choice model suggests that rational decision-making would increase the demand for human judgment, assuming that judgment is a normal good. A simple strategic model suggests that prediction errors arising from imperfect information can lead to war because future deterrence is not credible without patience. Since AI themselves cannot be patient, a human in the AI decision-making loop is necessary to keep the peace. If AI is this generation's nuclear arms race, then MAD is not the solution.

AI also cannot understand causality. Pearl (2009) notes the importance of theory and counterfactual reasoning for identifying causal relationships. AI can perhaps learn the counterfactual reasoning process. However, theory, i.e., imagination and conceptualization, is essential for starting the causal reasoning process in any practical sense. Theory lets humans see possible connections between things that may not exist in the sense of "is" but perhaps can exist as an idea in the sense of "could be" or even "should" be. This, then, is the foundation of judgment. Thus, giving autonomy to AI would perhaps lead to more accurate predictions and, in some cases, better outcomes, but it could also lead to a thought stasis in the sense of thinking of a future simply because AI cannot either conceptualize or imagine what is not already there—it is after all a prediction machine that depends on existing data. In that situation, quickly automated responses without any thought to potential future effects because of AI's lack of imagination could plunge the comity of nations into perpetual war.

Conclusion

Artificial intelligence has the potential to revolutionize the way we approach warfare. Rational choice models can help us understand how machines can assist in recognizing, interpreting, and predicting battlespace characteristics and improve

decision-making speed and accuracy. In this paper, I have focused mainly on two ideas emanating from the nature of AI. First, reducing the cost of AI driven by increased innovation will democratize its use among military powers and increase the demand for human judgment. Increasing the role of the human in AI decision-making is the rationally optimal thing to do. Second, the likelihood of prediction errors, even microscopically small, will reduce the credibility of deterrence. Therefore, thinking of an AI "arms race" in terms similar to a nuclear arms race is a mistake. This approach suggests that removing the human from the decision-making process, particularly the human capacity to imagine a moral future, could lead to more war. Therefore, we must keep human decision-makers in the loop to ensure that ethical and moral considerations are considered. It is, after all, the rational thing to do.

As a matter of policy, two possibilities emerge. AI should be leveraged to enhance prediction capabilities while humans focus on developing the moral capacity to judge. AI must also refocus on how we develop personnel strategy. An educated AI-ready armed force must be capable of deep patience and moral judgment. This means a humanistic education—particularly the old-fashioned kind of shorn of ideological scientism.

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